The chromosomal challenge of human embryos: prevalence of aneuploidy and mosaicism

Ilya Volodyaev, Ph.D.,[a,b,c] Anna Ivanova, M.Sc.,[a] Elena Korchivaia, M.Sc.,[a] Alexey Surnov, Ph.D.,[d] Ekaterina Pomerantseva, M.D., Ph.D.,[e] Igor N. Lebedev, M.D., Ph.D., Dr.Biol.Sc.,[f] Maria L. Semenova, Ph.D., Dr.Biol.Sc.,[a] and Ilya Mazunin, Ph.D.[c,g]

[a] Faculty of Biology, Department of Embryology, M.V. Lomonosov Moscow State University, Moscow, Russia; [b] Department of Reproductive Medicine, European Medical Center, Moscow, Russia; [c] Interdisciplinary Clinical Association for Reproductive Medicine, Moscow, Russia; [d] Research Division Type 1 Diabetes Immunology, Helmholtz Zentrum München, Munich, Germany; [e] Zentrum für Humangenetik Tübingen, Tübingen, Germany; [f] Research Institute of Medical Genetics, Tomsk National Research Medical Center of the Russian Academy of Sciences, Tomsk, Russia; and [g] Department of Biology and Genetics, Petrovsky Medical University, Moscow, Russia

Chromosomal aberrations, such as whole-chromosome aneuploidies, segmental aneuploidies, whole-chromosome mosaicism, and segmental mosaicism, are key factors influencing embryonic development and the outcomes of fertility programs. This analytic review critically examines the prevalence and origins of these genetic abnormalities, emphasizing the significant maternal contribution to whole-chromosome aneuploidies and the age-related nature of these aberrations. In contrast, segmental aneuploidies, whole-chromosome mosaicism, and segmental mosaicism appear largely age-independent and show considerable variability across studies, mainly due to technical artifacts and methodological differences. By analyzing the accumulated data, scrutinizing methodological discrepancies in preimplantation genetic testing for aneuploidies, and distinguishing between biologic phenomena and artifacts, this review aims to clarify the current understanding of chromosomal aberrations in human embryos and their impact on reproductive health. (F S Rev® 2025;6:100082. ©2024 by American Society for Reproductive Medicine.)

Key Words: PGT-A, mosaicism, aneuploidy, segmental aneuploidy, human embryo

ESSENTIAL POINTS

- Whole-chromosome aneuploidies are predominantly maternal in origin, increase in incidence by 6% yearly after the age of 35 years, and represent a major factor in the decline in reproductive outcomes with advancing maternal age.
- Segmental aneuploidies and chromosomal mosaicism are largely age- and whole-chromosome aneuploidies-independent and widely variable in reported prevalence (from 3% to >30%), mainly due to technical artifacts.
- The most reliable, meiotic segmental aneuploidies, show a clear two-phase developmental dynamic: 3%–10% in oocytes and supposedly 5%–20% in spermatozoa; 15%–25% in cleavage-stage embryos; and a decrease to 7%–15% in blastocysts.
- True biologic mosaicism likely affects 5%–15% of blastocysts, with its developmental dynamics still unclear, mainly due to technical artifacts, frequently indistinguishable from true biologic phenomena.
- Potential embryo self-correction still requires further investigation, and a clear separation of biologic features from technical artifacts is necessary to improve reproductive outcomes, underscoring the need for improved methodologies and standards.

Chromosomal aberrations are the primary cause of declining female fertility with age, as well as implantation failures and pregnancy losses (1–4). These aberrations encompass a wide range of events that disrupt the chromosomal structure—losses, gains, inversions, or rearrangements of whole chromosomes or their segments—and can affect the entire embryo or some of its cells. In the following, we review the latest knowledge on large-scale aberrations detectable by standard preimplantation genetic testing for aneuploidies (PGT-A) platforms, including next-generation sequencing (NGS), comparative genomic hybridization (CGH), and array CGH. Other types of genetic imbalances, such as translocations and point mutations, are beyond the scope of this article; we refer readers to recent reviews on these topics (5–7).

The four types of chromosomal aberrations discussed here are

Received June 26, 2024; revised November 10, 2024; accepted November 11, 2024.
I.V. and A.I. should be considered similar in author order.
Correspondence: Ilya Volodyaev, Ph.D., Department of Reproductive Medicine, European Medical Center, 15-1 Pravdy str., Moscow 125124, Russia (E-mail: ivolodyaev@gmail.com).

F S Rev® Vol. 6, No. 1, ■ 2025 2666-5719/$36.00
© 2024 American Society for Reproductive Medicine. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.
https://doi.org/10.1016/j.xfnr.2024.100082

whole-chromosome aneuploidies (WCA), segmental aneuploidies (SA), whole-chromosome mosaicism (WCM), and segmental mosaicism (SM). Whole-chromosome aneuploidies and SA involve all cells of the embryo, whereas WCM and SM affect only a subset of cells. Standard PGT-A profiles represent DNA content as a function of position along chromosomes, with WCA and WCM appearing as integer and fractional shifts across the chromosome, respectively (Fig. 1A and B). Segmental aneuploidies and SM are similar but occur in only a segment of a chromosome (Fig. 1C). Ideally, the measured DNA copy number in mosaic regions of the profile should indicate the percentage of cells with an aberrant karyotype. However, actual PGT-A profiles can contain various noises/artifacts (Fig. 1D), making profile interpretation ambiguous. Standard NGS platforms can detect segmental aberrations affecting at least 5–10 Mb, which will serve as our definition here. Smaller imbalances can be observed only with specific platforms, as in the study by Xie et al. (8), and should be treated separately (reviewed in the study by Picchetta et al. (9)).

These four types of chromosomal imbalances differ significantly in origin, including parental origin, meiotic or mitotic spindle errors, and cell cycle checkpoint failures. Thus, their incidence and distribution likely vary and should be analyzed separately, including potential associations between them. Technical artifacts and "noisy profiles" (Fig. 1D) further complicate the analysis and may introduce PGT errors: primarily false positives when artifacts are mistaken for aberrations but also false negatives when true mosaicism is missed amid background noise. In extreme cases of very noisy profiles or profiles obtained from poorly standardized platforms, the data can even be interpreted as multiple aneuploidies, also known as chaotic profiles. Such errors are a significant factor contributing to discordant results in repeat biopsies (10) and may also explain at least some cases of healthy live births reported after the transfer of embryos classified by PGT-A as mosaic (11, 12), abnormal (13, 14), or chaotic (15, 16). Importantly, the percentage of "noisy profiles" and the rates of mosaicism detected by PGT-A vary between laboratories, likely depending on operator experience and laboratory workflow (17–20). Thus, technical artifacts and interpretational issues of undefined frequency may be mixed with true biologic mosaicism and potential self-correction effects, with their clinical manifestations indistinguishable from each other.

Therefore, the collection and systematization of evidence-based data on the true frequency of chromosomal aberrations and the conditions of artifacts are of paramount importance. This critical review of the literature explores the prevalence and nature of these chromosomal abnormalities, focusing on their origins, dynamics during embryo development, and the estimation challenges posed by current diagnostic methodologies. By critically analyzing the latest data and distinguishing artifacts from biologic facts, we aim to provide a clearer understanding of these complex genetic phenomena, laying the groundwork for more informed and effective clinical practices in reproductive medicine.

WHOLE-CHROMOSOME ANEUPLOIDIES

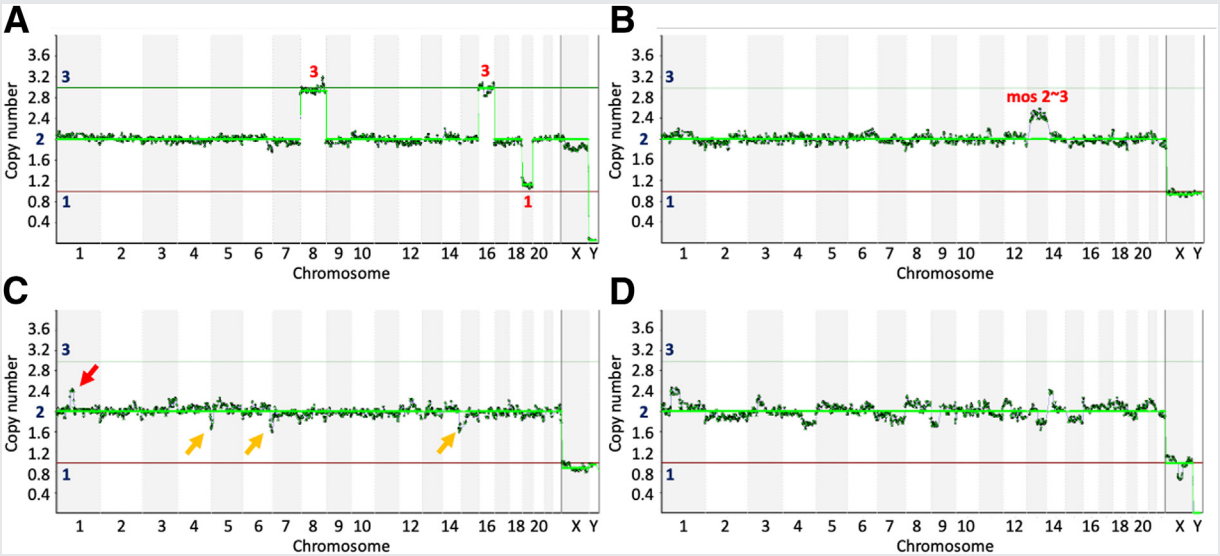
Concordance and reliability

As defined earlier, WCA is an imbalance affecting an entire chromosome in all cells of the embryo and is clinically diagnosed by integer abnormalities on the PGT-A profile (Fig. 1A). Extrapolating the results of a single trophectoderm (TE) biopsy to the entire embryo assumes a high degree of genetic homogeneity, which has been persistently criticized by some investigators (21–23). However, much of the evidence suggests that diagnoses of pure euploidy and WCAs through PGT-A are quite reliable. A summary of blastocyst rebiopsy-and-reanalysis data from 26 studies (10) showed 97% concordance for euploidy and 94% for WCA, which is a very good result considering the high heterogeneity of the original publications included in the review. Another effective (although ethically challenging and, therefore, rarely implemented) approach in these evaluations is to disclose PGT-A results after embryo transfer (ET) and compare them with clinical outcomes. A recent NGS-based study using this approach (24) reported zero false negatives for WCA: no pregnancies occurred after transferring 102 embryos with WCA, whereas 202 (65%) of 312 pregnancies followed the transfer of euploid embryos. Although cases of healthy live births after transfer of aneuploid (13, 14) and even multiple-aneuploid (chaotic) embryos (15) have also been documented, these appear to be sporadic. For accurate interpretation, the raw data should be carefully analyzed to distinguish artifacts and potential sample mix-ups from confined mosaicism and presumed self-correction. Prenatal testing in in vitro fertilization PGT-A programs (25) also showed very low PGT-A false negatives: just 1 of 1,139 cases. Thus, the modern version of PGT-A— TE biopsy and NGS testing—also called PGS 2.0, is highly reliable for assessing euploidy and WCA and is well suited for investigating the prevalence and origin of WCA as a biologic phenomenon.

Incidence, parental origin, and age relation

Preimplantation genetic testing–detected WCAs are strongly correlated with women's age, ranging from 20%–30% in women aged 25–35 years to 80%–90% at the ages of 42–44 years and beyond (26). After the age of 35, it shows a near-linear increase of approximately 6% per year (Fig. 2) (1, 26, 27). Importantly, this correlation aligns strongly with pregnancy and live birth rates when a woman's own oocytes are used, as opposed to in vitro fertilization cycles with donor eggs (Fig. 2A). This is considered the main reason for the well-known age-related decline in female fertility (1–3, 28, 29). Conversely, embryo aneuploidy has no proven correlation with the male partner's age, except for an indirect correlation through the age of the mother (30–33). Accordingly, the latest committee opinion from ASRM (American Society for Reproductive Medicine) (34) recommends PGT for women of advanced maternal age when using their own eggs, but not donor ones, and does not routinely recommend it for younger women or for any age of men.

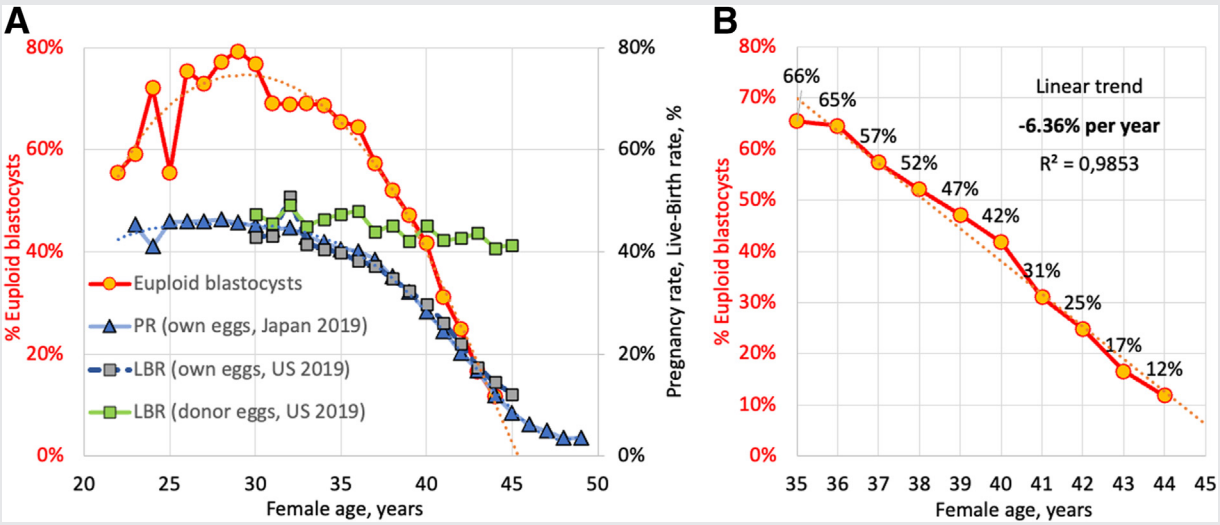
FIGURE 1



Examples of preimplantation genetic testing for aneuploidies with next-generation sequencing profiles: (**A**) aneuploid (trisomies 8 and 16 and monosomy 19); (**B**) mosaic trisomy 13 (corresponding to approximately 50% diploid cells and 50% cells with trisomy 13); (**C**) a profile with a questionable region in chromosome 1 (red arrow) that appeared similar to segmental mosaicism but could also be an artifact because there were other artifacts along the profile (marked with orange arrows); yet, the latter ones lie in the telomeric regions, exemplifying the so-called telomere effect, whereas the red arrow region does not; and (**D**) an uninterpretable ("noisy") profile.

Volodyaev. Chromosomal aberrations: prevalence. F S Rev 2025.

FIGURE 2



(**A**) Pregnancy rate (PR) with own oocytes (according to the Japan Society of Obstetrics and Gynecology records (1)), live birth rate (LBR) with own and donor oocytes (according to the US Fertility Clinic and National Summary Report (27)) in IVF cycles performed in 2019, and blastocyst euploidy rate assessed by quantitative polymerase chain reaction PGT-A (data from the study by Franasiak et al. (26)) vs. female age. Points corresponding to <50 cases per a 1-year range (PR data for the ages of 22 and ≥50 years and PGT-A data for the ages of ≥45 years) were excluded as unreliable. The fifth degree polynomial fits for PR, and PGT-A data sets are given as dashed lines. (**B**) Blastocyst euploidy rate from the left side, in the range of 35–45 years, with linear approximation (dashed line). The reasons for such a difference between the IVF effectiveness in the United States and Japanese reports (LBR in the United States practically equal to PR in Japan) are a separate issue that is not discussed here. Although NGS rather than PCR methods are currently the gold standard for PGT, the cited work (26) has a large sample size and high-quality data similar to the best NGS laboratories today. Therefore, we used it to estimate the basic prevalence of aneuploidies, rather than newer publications, to avoid discussing mosaicism and related artifacts that have become particularly relevant with the introduction of NGS. IVF = in vitro fertilization; LBR = live birth rate; NGS = next-generation sequencing; PCR = polymerase chain reaction; PGT-A = preimplantation genetic testing for aneuploidy; PR = pregnancy rate.

Volodyaev. Chromosomal aberrations: prevalence. F S Rev 2025.

As can be inferred from these data, most embryo aneuploidies are of maternal origin, a conclusion supported by PGT-A with single nucleotide polymorphism array (33, 35, 36) and by the reported sperm aneuploidy rates (37). For instance, in a study of 31,228 donor sperm samples, only 920 cells (2.9%) were aneuploid (37). Although this cannot be directly applied to embryos, it roughly corresponds to 10% paternal contribution of all embryo aneuploidies for a female partner aged 30–35 years (approximately 3% sperm aneuploidy / 30% embryo aneuploidy) and only 5% for the age of 40 years (3% sperm aneuploidy / 60% embryo aneuploidy). This is an upper estimate of paternal contribution, assuming an equal probability of fertilization for euploid and aneuploid sperm, which is unlikely; therefore the actual contribution could be even lower.

Studies on PGT-A directly assessing parental haplotypes yield similar estimates. At the blastocyst stage (among TE biopsies), maternal aneuploidies were reported to be 82% (335 of 408) in one study (35) and 85% (3,522 of 4,156) in another (33), with an even higher proportion among whole-chromosome abnormalities: 90% (319 of 354) (35).

At the same time, cleavage-stage WCA estimates reveal some discrepancies. An earlier study by the same group (36) estimated maternal aneuploidies at only 69%, with 90% for trisomies (405 of 451) and (strangely) only 51% for monosomies (275 of 540). They also reported an unusually high total aneuploidy rate of 72% (the average female age was 38 years; compare to Fig. 1) and a significant number of biparental aberrations (32%). Biologically, this pattern would imply the following: a high number of mitotic errors at early cleavage selectively enriching embryos with paternal aneuploidies and their further intense correction before the blastocyst stage. This would also require an extensive number of arrested embryos, first selectively with maternal errors and then with paternal ones, for which there is no evidence. Thus, the investigators' data are more likely explained by technical artifacts, such as chromosomal loss or DNA degradation, resulting in overestimated aneuploidy, including monosomies and biparental imbalances.

Notably, a subsequent study from the same group (33) corrected their assessment to 84% maternal trisomies (1,695 of 2,013) and 77% monosomies (2,088 of 2,720), in better agreement with the aforementioned blastocyst stage data. Moreover, after excluding all biparental abnormalities (which have a higher risk of artifacts), the estimate for maternal aneuploidies was 83.5%, consistent with other publications.

Overall, meiotic errors in oogenesis are the primary cause of chromosomal abnormalities in embryos, far surpassing other factors. These errors result in WCA at all embryo stages, although they can be mimicked by further mitotic errors or artifacts (discussed in the following).

Origin: meiotic or mitotic?

Although meiotic errors typically lead to complete aneuploidies and mitotic errors to mosaic ones, the actual pattern is often more complex. For example, very early *mitotic* errors may result in a completely aneuploid embryo if some euploid daughter cells are lost. Late mitotic errors may affect too few cells and fall below the detection limits for mosaicism (typically 20%–80% (38, 39)). Moreover, a mitotic error in some cells of an aneuploid embryo may occasionally restore euploidy through mechanisms such as chromosomal loss (for trisomy) or acquiring uniparental disomy (for monosomy), leading to an embryo classified as mosaic, with the initial meiotic error overlooked (40). Such patterns are also discussed in the context of embryo self-correction hypothesis (41), which, however, is challenging to distinguish from technical artifacts. Thereby, both full and mosaic aneuploidies observed in clinical practice may originate from either meiotic or mitotic errors.

An elegant way to distinguish true meiotic aneuploidies is to assess chromosome haplotypes in cases of trisomy (33, 36). If all three copies of the trisomic chromosome have different haplotypes (a pattern called "both parental homologues" [BPH]), it indicates that one gamete carried two chromosome homologues, suggesting nondisjunction in meiosis. However, if two of the three copies are identical, this does not rule out a meiotic error because it could still result from meiotic errors without recombination. Thus, the percentage of three-haplotype trisomies is a *lower estimate* of meiotic errors, whereas their actual frequency may be higher.

For maternally derived trisomies, the BPH pattern was observed in 69% of cleavage-stage embryos (1,164 of 1,695) and 64% of blastocysts (1,031 of 1,606) (33), indicating that roughly two-thirds of maternal aneuploidies are of proven meiotic origin. The remaining one-third may be either meiotic or mitotic in origin, which cannot be distinguished from these data alone. For paternally derived trisomies, the BPH percentage was 14% at both cleavage (41 of 318 embryos) and blastocyst (45 of 321 embryos) stages, suggesting that meiotic origins are much rarer for paternal aneuploidies than for maternal ones.

However, certain inconsistencies in these estimates should be noted. Although sperm aneuploidy rates are 2%–3%, the rate of paternal or biparental aneuploidies among embryos is several times higher. This discrepancy may result from inaccurate estimates, DNA degradation, or chromosome loss; however, a significant portion likely arises from early mitotic errors, primarily from the first mitosis. Indeed, the proportion of zygotes formed from fertilizing an aneuploid oocyte with an aneuploid sperm can be roughly estimated as 3% (aneuploid sperm) times 30%–90% (aneuploid oocytes, depending on the female age), that is, 1%–3% of all zygotes. Although this is a rough calculation, it likely overestimates the actual number because the probability of fertilization is likely lower for aneuploid compared with euploid gametes. Thus, a major portion of biparental aneuploidies may originate from early mitotic errors rather than from aneuploid-aneuploid fertilization. Altogether, this may account for approximately 5%–15% of aneuploidies and manifest in non-BPH trisomies and most biparental aneuploidies.

When considering parental origin, pure meiotic (BPH) aneuploidies are even more maternally skewed compared with general estimates: for example, 96.6% at the cleavage-stage (1,164 of 1,205) (33) and 95.8% (1,031 of 1,076) (33) or 91.9% (137 of 149) (35) for blastocysts. Maternal-derived trisomies follow the same age-related pattern as general

maternal aneuploidies. This difference (80%–85% maternal aneuploidies in general vs. >90%–95% maternal BPH aneuploidies) is expected because total aneuploidies include both BPH- and non-BPH-manifesting meiotic, as well as mitotic ones, which can affect maternal and paternal chromosomes with similar probability.

In summary, the distribution of all aneuploid embryos is roughly as follows:

- At least 60% (likely more, age-dependently) have oocyte-derived meiotic aberrations.
- Two to three percent have sperm-derived meiotic aberrations.
- Roughly 25% contain maternal imbalances, and 10%–15% contain paternal imbalances of uncertain origin (either meiotic or mitotic). A significant portion likely arises from the first mitosis (zygote, 2-cell), whereas others may still be meiotic, although the exact proportion is unknown.

Of the first two categories (proven meiotic aneuploidies), most occurred during meiosis I: 70% (35); 80% (42); and from 60% to 100% depending on the chromosome number (43). Thus, WCAs are predominantly maternal and meiotic in origin, and they are the root cause of the well-known decline in all reproductive outcomes with advancing maternal age. The average euploidy rate in oocytes is approximately 30% for women aged 25–35 years and declines by approximately 6% per year thereafter.

SEGMENTAL ANEUPLOIDIES

Estimates and relation to age and WCA

Unlike WCA rates, which are consistent across studies using comparable platforms, results for SAs vary significantly: from approximately 3% at the blastocyst stage in Colorado Center for Reproductive Medicine (44) to 30% in other studies (45, 46) or even higher. These discrepancies between laboratories likely arise from differences in sample preparation and transport, as well as in bioinformatics and analytic methods. The first issue can result in artifactual aneuploidies because of DNA degradation and loss, whereas the second may lead to false-positive and false-negative results, depending on the algorithm's parameters. Furthermore, even within the same laboratory, low concordance (approximately 40%–50%) is observed in multifocal biopsy of embryos initially diagnosed with SA (47). Such discrepancies can be attributed to SA confined to a specific area of the embryo or, alternatively, to artifacts. Therefore, the observed SA should be considered "apparent," implying a combination of both artifacts and biologic phenomena, the ratio of which is yet to be precisely determined.

Notably, despite the variability in absolute SA estimates, all available data agree that SA is age-independent (32, 35, 44, 45, 48, 49) and shows no discernible correlation with WCA (Appendices A and B and Supplemental Tables 1–8, available online). Thus, SA and WCA can be considered independent phenomena.

On average, the probability of SA is linearly correlated with chromosome size and is higher for (longer) q-arms than for (shorter) p-arms (42, 44, 48). This finding is consistent with its independence from WCA and its partially artifactual origin. Additionally, specific SA "hotspots" have been identified where the probability of SA occurrence is much higher than in the other parts of the genome (44). These "hotspots" are primarily located at chromosomal distal ends.

Parental origin and dynamic in embryo development

Although WCA is overwhelmingly maternal, the situation with SA appears quite different. For instance, paternal and maternal SAs have been estimated at 70% and 30% (35) or 67% and 33% (44), respectively, which contrast sharply with WCAs, estimated at 6% paternal, 88% maternal, and 6% biparental in the same study (44). It is important to note that both studies refer to the "total observed SAs" at the blastocyst stage and cannot distinguish between parentally inherited, de novo, and artifactual SAs. Additionally, the sample sizes were relatively small: 54 (35) and 21 (44) SA-affected embryos. Nonetheless, the parental contribution to observed SA at the blastocyst stage can be roughly estimated as two-thirds paternal and one-third maternal.

A separate group of studies compared the SA rates at different developmental stages (Table 1) (45, 50, 51). Notably, although the absolute SA estimates varied substantially between studies, all reported a distinct 2-phase dynamic: twofold to threefold increase in SA level from oocytes to the cleavage-stage and 1.5–2-fold decrease from the cleavage to the blastocysts stage. The question is whether this dynamic is real or due to artifacts. Potential artifacts could include the following: artifactual SAs when working with single DNA copies, particularly in oocytes and cleavage-stage embryos; missed SAs in individual cells at the blastocyst stage due to averaging with other cells in the biopsy; and various artifacts affecting all data equally across stages. None of these factors can account for the increase in SAs from oocytes to cleavage-stage because both stages involve biopsies containing a single DNA copy. Thus, this increase is likely a true biologic phenomenon. Regarding the decrease in SA from cleavage to blastocyst, although it could potentially be explained by missed SAs, this would require excessive heterogeneity in SA presence across the embryo, which has not been observed in any related studies (see reviews (41, 52–54)). Therefore, the decrease is also likely real.

What may explain this dynamic biologically? At fertilization, oocyte SAs are augmented by sperm-derived ones, which could be assessed by either investigating the sperm SAs or comparing the SA rates in oocytes and zygotes. Few and inconsistent estimates of sperm-derived SAs are available (see discussion (55)), and zygote-stage genetic data cannot be obtained for ethical reasons. Thus, sperm- and oocyte-derived SAs can only be estimated indirectly. De novo mitotic SAs and any artifacts likely affect maternal and paternal chromosomes equally (42). Therefore, the proportion between paternal and maternal SAs at the zygote-stage is likely not enhanced but rather smoothed over development. Thus, the two-third parental vs. one-third maternal SAs observed at the blastocyst stage can be used as a *lower estimate* of the initial proportion, suggesting that sperm-derived SAs are

TABLE 1

Prevalence of segmental aneuploidies in human oocytes and embryos, according to different studies.

		N samples			SA rate, %		
Study	**Protocol**	**Oocytes**	**Cleavage**	**Blastocyst**	**Oocytes**	**Cleavage**	**Blastocyst**
Fragouli et al. (50)	Cytochip aCGH	420	754	1,046	4.3%	15.3%	7.0%
Babariya et al. (45)	Cytochip aCGH	452	1,762	1,327	10.4%	24.3%	15.6%
Magli et al. (51)	Cytochip aCGH	331	886	570	6.3%	16.6%	11.2%

Note: aCGH = array comparative genomic hybridization; SA = segmental aneuploidies.

Volodyaev. Chromosomal aberrations: prevalence. F S Rev 2025.

approximately twice as frequent as oocyte-derived ones. This matches well with the observed SA increases from the oocyte to the cleavage-stage and is likely its main cause.

Embryo development may involve three divergent processes: the appearance of de novo mitotic SAs as cells divide; partial elimination of SAs if SA-affected embryos are more likely to arrest than fully euploid ones; and hypothetical SA correction. The observed *decrease* in SAs from cleavage to blastocyst suggests that the second and third processes are at play, with de novo SAs as a second-order factor. Distinguishing between the two requires a comparison of SA levels and their parental origin in both developing and arrested embryos at different stages using the same methodology, which is currently lacking in the literature. Instead one should infer their contribution indirectly from SA data observed in developing embryos (50).

The contribution of SA elimination can be roughly estimated as follows. On average, approximately 60% of cleavage-stage embryos progress to the blastocyst stage (56, 57). If this is accompanied by an α-fold decrease in the SA rates, the SA-affected embryos should be α times less likely to develop to the blastocyst stage compared with the average:

$$\alpha \cdot n_2 / N_2 = n_1/N_1 \Rightarrow \alpha \cdot n_2/n_1 = N_2/N_1$$

where N_i is the total number of viable embryos, n_i is the number of viable SA-affected embryos, and n_i/N_i is the SA rate at time point i. Whether this is sufficient to explain the cleavage to blastocyst SA decrease, or whether it requires additional "SA correction," can be judged by comparing the SA rates among cleavage-stage embryos, blastocysts, and embryos halted between these two stages and calculating the α. If it appears significantly less than the observed 1.5–2, one will get strong arguments in favor of the "SA correction" hypothesis; otherwise, it should be left aside according to the Occam razor. However, these simple data are not yet available.

Interestingly, although the SA rates are independent of WCA and the patient age, they have been found to be correlated with the quality of the blastocyst inner cell mass (ICM) and TE: 1.7%, 6.4%, and 13.2% for TE grades A, B, and C and 4.6%, 7.4%, and 13.1% for ICM grades A, B, and C, respectively (58) (the embryo numbers were as follows: TE_A, 59; TE_B, 1,097; TE_C, 665; ICM_A, 174; ICM_B, 1,142; ICM_C, 504; and total, 1,821). This correlation aligns with the lower developmental potential of SA-affected embryos, mentioned earlier. Although the association with TE quality may be influenced by a higher risk of artifacts in low-quality biopsy samples, this does not apply to ICM quality, suggesting a real negative correlation between SA and embryo developmental potential.

Origin: meiotic, mitotic, or artifacts?

Unlike WCAs, which primarily originate from maternal meiosis, SAs appear to have a different origin. To assess it, studies often use multifocal biopsies (typically dividing the whole embryo into several pieces) (48) or label all cells within the embryo (45). The uniform presence of a certain SA is interpreted as meiotic origin, whereas mosaic presence in some biopsy samples is noted as mitotic. However, this may be biased by artifacts. For example, if only one of several biopsies from the same embryo contains a particular imbalance, does that indicate mosaicism or an artifact because of DNA degradation? As there are no definitive criteria to distinguish these scenarios, the following conclusions can be drawn from available data (Supplemental Fig. 1, available online; see also the study by Girardi et al. (48)):

- If a specific nonmosaic aneuploidy is uniformly present in all biopsy samples, it is most likely true and meiotic.
- If it is present in all samples in mosaic form, it is likely true and mitotic, unless all PGT profiles are noisy and uncertain.
- If it is present in two or more (but not all) biopsy samples, it is likely true and mitotic.
- If it is present in only one biopsy sample, it may be either mitotic or an artifact, which cannot be distinguished definitely. If it is clearly nonmosaic and/or if the rest of the profile is even, it may be a true mitotic aneuploidy.

On the basis of these criteria, the following data can be derived from the available studies (Table 2) (44–46, 48, 51, 59, 60). The most reliable option is surely the "likely meiotic" SAs, whose rates appear fairly consistent across studies: approximately 5%–10% of all embryos, independent of WCA (Table 2). Calculating the mitotic SA rates yields large discrepancies, from <0.2% (44) to 9%–12% (46, 59), with even greater variation for "questionable SAs." These discrepancies are due to the difficulty in distinguishing true mitotic SAs from artifacts and the limitations in assessing localized SAs using multifocal biopsies. Plotting the SA rate against SA concordance across studies shows an inverse relationship: the higher the reported SA rate, the lower the concordance achieved in multifocal biopsies

TABLE 2

Summary and reanalysis of data on segmental aneuploidies in human blastocysts.

						% Embryos among all, affected by[a]		
Study	**Protocol**	**N embryos, stage**	**Total SA rate**	**N dissected embryos with SA**	**N parts, each embryo was divided into**	**Likely meiotic SAs**	**Likely mitotic SAs**	**Questionable SAs**
Babariya et al. (45)	PGTseq-A Illumina	1,762 Cleavage	24.3%	8	Whole embryo (targeted FISH reanalysis)	9.1%	6.1%	9.1%
		1,327 Blastocyst	15.6%	7		2.2%	11.1%	2.2%
Kim et al. (59)	PGTseq-A Illumina	300 Blastocyst	25%	75	5 (1 initial TE biopsy + 4 after donation)	10.5%	8.6%	5.9%
			28% only mosaic SA	85		—	11.8%	16.2%
Navratil et al. (46)[b]	VeriSeq Illumina	89 Blastocyst	29%	89	3	6.4%	5.8%	16.8%
Girardi et al. (48)	Ion Reproseq PGS Thermo Fisher Scientific	8,137 Blastocyst	5.6%	53	5	1.8%	1.7%	2.2%
McCarty et al. (44)	VeriSeq Illumina	89,226 Blastocyst	5.2%[c]	54	3 (2 TE biopsies and the rest of the embryo)	5.0%	0.2%	
Magli et al. (51)	Cytochip aCGH Affymetrix	570 Blastocyst	11.2%[d]	10	Whole embryos	7.8%	3.4%	

Note: However, the studies mostly did not separate interstitial deletions/duplications from terminal 2-segment rearrangements, which may result from segregation of balanced chromosomal aberrations and, thus, should not be included in analyses of meiotic or mitotic origin (60). FISH = fluorescence in situ hybridization; SA = segmental aneuploidies; TE = trophectoderm; WCA = whole-chromosome aneuploidies.

[a] Recalculated from the investigators' data as follows: (% of embryos with SA) × (% of SAs with the corresponding interbiopsy concordance).

[b] Recalculated from Supplemental Table 1 in the study by Navratil et al.(46).

[c] The investigators classified all embryos into "euploid," "aneuploid," and "segmentally aneuploid"; thus the reported 3.1% SA rate was the rate of SA+ WCA− embryos among all. Because SA incidence was likely independent of WCA, the true SA rate will be the reported 3.1% divided by the euploidy rate in the study, that is, approximately 5.2%.

[d] The SA rate was found to be 11.2% (64 blastocysts of 570 TEs biopsied) with a total of 77 SA events. Most TE biopsies had single SAs (53 biopsies) either as the only abnormality (24 of 53) or in combination with WCA (29 of 53).

Volodyaev. Chromosomal aberrations: prevalence. F S Rev 2025.

(Supplemental Fig. 2). This suggests that a significant proportion of observed SAs are artifacts, although they cannot be quantified from these data alone.

Interestingly, nearly two-thirds of detected SAs are deletions, whereas the remainder are duplications: 1,872 vs. 939 (44) and approximately 16,500 vs. approximately 10,000 (49). This is also consistent with their partially artifactual origin because artifacts are more likely to involve DNA loss rather than excess.

It is important to distinguish between terminal and interstitial deletions/duplications. Among terminal imbalances, 2-segment imbalances are usually inherited rather than de novo and originate from abnormal meiotic segregation of a parental reciprocal translocation. Such an SA should be present in balanced form in the parent's genome and become unbalanced during the quadrivalent meiotic segregation, resulting in terminal rearrangements on different chromosomes. Therefore, two-segment deletions and duplications on different chromosomes should not be treated the same as interstitial or single terminal imbalances, which may be inherited or de novo. We recommend focusing on interstitial and single-segment terminal imbalances, excluding such patterns, when assessing SA origin (60). However, most published data lack this level of detail, making it impossible to distinguish these patterns (Table 2). Thus, the published SA rates should be considered *upper* estimates. Where noted, the proportion of two- and three-segment SAs ranged from 19% to 30% of the total, suggesting that the SA estimates above may be overstated by about a quarter (45, 51).

In conclusion, SAs are generally independent of WCA and age; however, diverse estimates from different laboratories include numerous artifacts yet to be properly analyzed and excluded. At the same time, SAs are associated with poor embryo morphology and arrest, which is likely the main reason for the decrease in SA during embryo development. The bottom-line estimate of the real biologic SAs is approximately 5% at the blastocyst stage, and the true rate remains to be elucidated.

MOSAIC ABERRATIONS

Estimates and relation to age and WCA

Similar to SA and unlike WCA, mosaicism estimates vary greatly across published studies: from <3% consistently reported by Colorado Center for Reproductive Medicine (61, 62) to 30%–35% (63–65) and even higher (66, 67). This discrepancy points to assessment issues, including artifacts from sample preparation or transport, and false positives or negatives from bioinformatics and analytic methods. Notably, even studies using the same or similar NGS platforms demonstrate this heterogeneity. For instance, most of the aforementioned publications used either Illumina, AB Vector (which have near 100% concordance (68)), or FAST-SeqS (which has 98% concordance with Illumina (49)). Thus, the reasons for such diverse estimates cannot be eliminated by simply standardizing platforms and require a more in-depth investigation. Similar to SA, multifocal biopsies show low concordance between embryo samples, initially diagnosed as mosaic: for example, 43% (10) and even 27% (46). As with SA, the observed mosaicism should be labeled as "apparent," suggesting that both biologic phenomena and artifacts are at play.

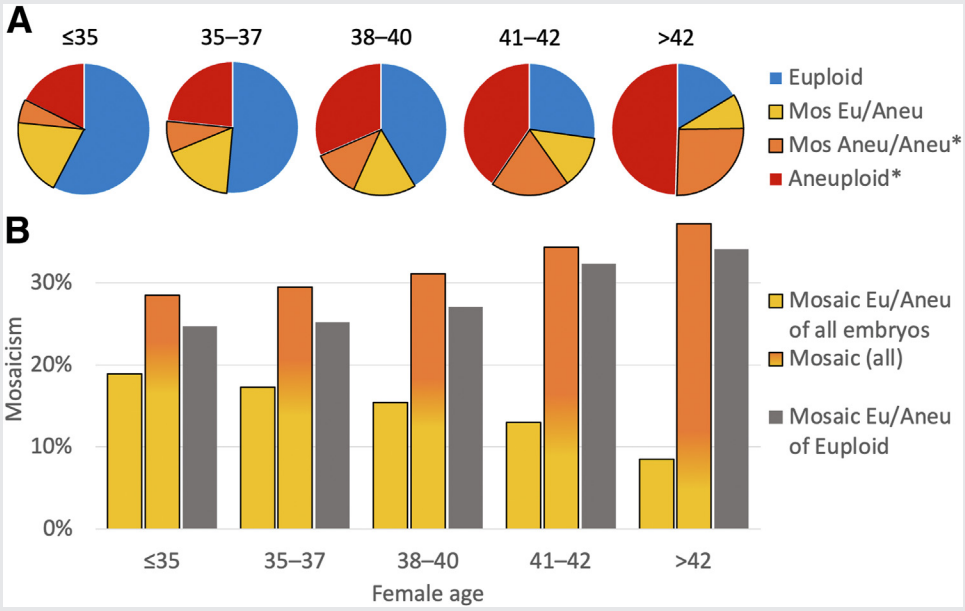
Importantly, despite the variability in absolute estimates, most available data agree that WCM is largely age-independent (63, 65, 69). However, some investigators report a decrease in the WCM rate with age, such as from 18.9% in women aged <35 years to 8.5% in women aged >42 years (38), from 18.1% in women aged <35 years to 9.3% in women aged >35 years (70), from 14.2% in women aged 20–24 years to 5.3% in women aged >40 years (49). This discrepancy arises from differences in how the mosaicism percentage is calculated. Some investigators categorize embryos as either euploid, aneuploid, or mosaic, mixing aneuploid-aneuploid mosaics with full aneuploid embryos (38, 49, 70). They, thus, calculate the percentage of mosaic-only (WCM+ WCA−) embryos relative to all embryos, missing WCM+ WCA+ embryos and understating the WCM rate. To accurately estimate WCM, it should be calculated across all embryos regardless of other factors or separately for different subgroups, such as mosaic-only among otherwise euploid or WCM+ WCA+ among all WCA+ embryos. When recalculating the data from the study by Armstrong et al. (38) this way, the mosaicism rates either remain stable or slowly increase with age (Fig. 3A and B) (38), a trend also found at recalculating (49, 70). Thus, the WCM rate does not decrease with age, which aligns with estimates in other studies (63, 65, 69). Moreover, there seems to be a slight increase in WCM, likely corresponding to the known gradual decline in mitotic molecular machinery (71, 72).

Interestingly, the mosaicism rates among non-WCA embryos are slightly lower than those among WCA embryos (65) or among all embryos (38), suggesting that observed ("apparent") mosaicism is more likely to occur among aneuploid embryos. Because "apparent WCM" includes both biologic phenomena and possible artifacts, these data alone cannot determine whether true mosaicism is associated with WCA. It would be logical to expect a weak association between WCA and WCM because of the overlap in their biologic regulatory networks (71–74) or because gene expression disruption in aneuploid embryos may affect mitosis-related genes (65). However, these hypotheses remain unproven and can only be considered if the WCA-WCM association is eventually confirmed. Because the differences between WCA, non-WCA strata, and all embryos are relatively small (<15%) and the WCA-WCM association is quite weak (e.g., $K_a = 0.140$ (65)), observed mosaicism can be considered independent of aneuploidy as a first approximation (Appendix C and Supplemental Tables 9–15).

Origin, dynamic in embryo development and estimation issues

First of all, mosaicism results from mitotic errors, regardless of whether there was an initial meiotic error, and is likely to affect chromosomes from both parents equally. It probably originates from the first mitotic divisions, known to be more error-prone (75, 76). Furthermore, mosaicism appears to be largely independent of maternal age and embryo ploidy:

FIGURE 3



(**A**) Euploidy, aneuploidy, and mosaicism rates for different age groups (38). The aneuploidy and aneuploid-aneuploid mosaicism rates were estimated from the data in the original paper. (**B**) Mosaicism rates calculated by different algorithms from the data in **A**: yellow represents the originally reported "clinical mosaicism rate" calculated as the number of euploid/aneuploid mosaics divided by the total embryo number (the yellow sector in **A**). Yellow-orange, the real mosaicism rate, the total number of mosaic embryos divided by the total embryo number (the yellow and orange sectors in **A**). Grey, mosaicism rate among otherwise euploid embryos, the number of euploid/aneuploid mosaics divided by the total number of otherwise euploid embryos (i.e., the ratio of the yellow sector to the sum of the yellow and blue sectors in **A**).

Volodyaev. Chromosomal aberrations: prevalence. F S Rev 2025.

its rates differ insignificantly between euploid and aneuploid embryos (38, 39) (Supplemental Tables 9–15).

However, estimating true mosaicism levels and its dynamics using embryonic development faces major challenges. First, cleavage-stage biopsy is now well-known to impair subsequent embryo development and implantation (77) and is, therefore, rarely used (unlike TE biopsy, which does not have a negative effect (34)). Mosaicism assessment requires sampling several cells from the same embryo, which is even more invasive. As such, cleavage-stage mosaicism assessment is mostly based on earlier studies using the fluorescence in situ hybridization (FISH) method or on several small sample studies involving rejected embryos.

Second, FISH and the currently standard NGS are fundamentally different methods with artifacts that cannot be recalculated into each other. While NGS and array CGH artifacts are related to DNA degradation and uneven amplification, FISH is prone to "losing chromosomes," which may increase the apparent level of mosaicism (a direct comparison of methods is shown in Supplemental Table 16 and in the studies by Treff et al. (78) and Velilla et al. (79)). The FISH data typically analyzed several chromosomes, usually 4–10 chromosomes, whereas both aneuploidy and mosaicism can affect all chromosomes (80). Thus, FISH-based mosaicism estimates across all chromosomes would be close to 100%.

Overall, realistic mosaicism levels have only been achieved in recent NGS-based studies which still contain some artifacts. Most cleavage-stage mosaicism estimates are FISH-based, making direct comparisons difficult. Thus, assessing mosaicism dynamics across developmental stages relies on FISH-based studies, which tend to overestimate aneuploidy.

Here are the examples of studies that directly compared mosaicism at different stages within a single research (note small sample sizes; Supplemental Table 16):

- 15.2% in 2–4-cell stage (59 embryos), 49.4% in 5–8-cell stage (81 embryos), 58.1% in morulas (43 embryos), and 90.9% in day 6 blastocysts (33 embryos); method: FISH for nine chromosomes (all but blastocysts) and FISH for five chromosomes (blastocysts) (81)
- 69% in day 2–3 4–10-cell stage embryos (16 embryos), 86% in day 4 morulas (14 embryos), 80% in day 5 early blastocysts (5 embryos), and 87% in day 5 expanded blastocysts (15 embryos); method: FISH for five chromosomes (67)
- 50% in day 3 embryos (biopsy of 2 blastomeres, 121 embryos), 45% in day 5 blastocysts (49 embryos), and 65% in embryos arrested on days 3–4 (27 embryos); method: FISH for 10 chromosomes (82)

Combining data from different publications (Supplemental Table 16), the following FISH-based mosaicism estimates can be obtained: 44% (day 3); 55% (day 4); 61% (blastocysts);

64% (day 3 arrest); and 68% (day 4 arrest). Despite the view that mosaicism may peak at the cleavage-stage and decrease thereafter, these data lack strong evidence because of technical artifacts and an absence of high-quality contemporary studies that directly assess mosaicism at different stages.

Altogether, mosaicism appears largely independent of age and WCA (or slightly positively correlated with both). Its estimates vary widely across laboratories due to numerous artifacts that need further analysis and exclusion. Current genetic methods involve fundamentally different artifacts, leading to significant variability between studies. True biologic mosaicism likely affects <15% of embryos, with a lower estimate of approximately 5% (if regarded independently of other factors).

ARTIFACTS VS. SELF-CORRECTION

Several studies have examined the outcomes of ET identified by PGT-A as abnormal. Most of these cases involved mosaic embryos (11, 12, 83–87); however, some were fully aneuploid (13, 14) and even multiple-aneuploid (chaotic) (15, 16). Understandably, the overall pregnancy and live birth rates were lower in these cases (e.g., 45% of 2,031 mosaic ETs vs. 59.1% of 14,673 euploid ETs (11)), and the miscarriage rates were higher (22.2% of 914 pregnancies vs. 8.9% of 8,672 pregnancies, correspondingly (11)). However, when an ongoing pregnancy was obtained, prenatal testing showed that the fetus was karyotypically normal in a vast majority of cases (98.8% (11)), and the newborns showed no significant difference from those after euploid ET (e.g., just 1 case of overt gross abnormalities out of 498 newborns (11)).

Notably, the level of mosaicism, rather than the specific chromosomes involved, is the primary predictor of implantation and developmental potential. Low-level mosaics are practically indistinguishable from euploid embryos, whereas the potential of high-level (87, 88) or complex abnormal (83, 89, 90) mosaics is significantly reduced. There was also no significant difference in pregnancy outcomes between mosaic monosomies and trisomies (83, 89–91), one or two affected chromosomes (89, 90), or their size (89). These clinical outcomes may be explained by either technical factors, such as sample mix-ups, artifacts, and data misinterpretation, or biologic factors including mosaicism confined to TE or its spots, preferential arrest of aneuploid cells during embryo development, or self-correction by a second mitotic error restoring euploid chromosomal set.

As discussed earlier, the observed decline in the SA rate from the cleavage to blastocyst stage is likely real, possibly due to the preferential development of euploid embryos and/or cells. A similar pattern is suggested for WCM, although unproven due to the lack of good-quality cleavage-stage data, making it only speculative at this point. Aneuploidy disrupts normal gene expression, and although these disruptions vary, aneuploid embryos are generally more likely to arrest (64, 65, 92–94). This pattern likely applies to cells as well. Model studies have shown that aneuploid cells are more prone to developmental arrest than euploid cells (95, 96). Additionally, a recent study demonstrated that karyotypically abnormal cells tend to be excluded from the blastocyst or located in the trophoblast, away from the fetus (97). These findings may support the self-correction hypothesis thoroughly discussed in a number of recent papers (41, 98).

The role of technical factors is evident from the variation in mosaicism and SA reports across laboratories. It is also supported by the study by Rodrigo et al. (16), which reported approximately 40% of embryos initially identified by NGS as chaotic to be euploid on repeated analysis, indicating that the primary diagnosis was likely due to misinterpretation of noisy profiles. Factors that contribute to artifacts (also known as technical mosaicism) are not fully understood and depend largely on the methods used. The main sources of artifacts in NGS, array CGH, and other whole-genome methods are DNA degradation and uneven amplification, whereas artifacts in FISH mainly involve chromosome loss or false signals. Importantly, the risk of artifacts in NGS is the greatest when analyzing a single cell and decreases as the number of cells in the sample increases, whereas the opposite is true for FISH. This makes mosaicism estimates from these two methods incomparable, limiting any reliable analysis to data obtained using the same method.

To reduce the risk of artifacts in NGS, methodical recommendations for the embryological step include analyzing as many undamaged cells as possible, washing the sample in multiple buffer drops, tubing samples as soon as possible after biopsy, and (perhaps most importantly) maintaining the cold chain after tubing and freezing the sample (99–104). The genetic step of NGS involves whole-genome amplification, which should be as homogeneous as possible and requires specially designed primers. In addition, because its homogeneity is in any case compromised, for example, by extremely GC-rich regions, it has to be further bioinformatically corrected, making the combination of primers and correction algorithms the main value of PGT platforms. Thus, here, the first step toward minimizing these artifacts is the use of standardized and validated platforms, which may reduce the artifacts associated with nonstandardized methods. However, even standard platforms do not eliminate all technical issues and misinterpretations.

A major problem of all PGT-a platforms is very low genome coverage. The coverage depth (the average number of times a base is sequenced) in PGT is mostly 0.1× (104), as opposed to 20×–30× in standard medical sequencing (105). However, meaningful increases in this parameter require too high amplification from the original single DNA copies and are currently too expensive and are associated with systematic bias of nonuniform amplification. Thus, the only thing left to do is to keep the genome coverage at the accessible level. Another primarily important issue is the signal "noisiness," partially determined by the low coverage (because there are not enough points to smooth sporadic outbursts) and systematic bias in sequencing homogeneity. The most common quality parameters are derivative log ratio (DLR) (106–108), coefficient of variation, and median noise (68), all reflecting the profile dispersion around its mean or median values, separately determined within each chromosome or its segment (68). The DLR parameter originates from array CGH, where the tested genome is

compared with the reference sample, and the signal noise is best estimated by fluctuations in their ratio in logarithmic scale (60). In NGS, in the absence of reference, DLR is calculated differently (108) but generally retains the meaning of the degree of signal noise (106, 107) or replaced with other noise estimates (68).

Altogether, such integrative noise characteristics play a primary role in PGT-A analysis, particularly in identifying aneuploidy and artifactual mosaicism. Lower values for these parameters (typically a DLR of <0.2) indicate reduced background noise and, consequently, more confidence in detecting true chromosomal abnormalities. Higher values (typically a DLR of >0.4) indicate greater technical noise or worse-quality whole-genome amplification and mean less ability to distinguish between artifacts and real aneuploidies, especially in mosaicism detection. Further progress in PGT-A is actually a task of minimizing noise metrics, at both embryological and genetic/bioinformatics stages.

More advanced computational approaches have been developed using convolutional neural networks (109) or hidden Markov models (110) trained on animal data. Despite an evident progress in machine learning, these methods still require careful validation in the context of PGT-A.

Altogether, a thorough and comprehensive consideration of technical issues surrounding PGT is essential to minimize artifacts, reduce false results, and maximize the reliability of the method. All of these have yet to be collated, scrutinized, and critically analyzed, which is a separate comprehensive topic beyond the scope of this review.

CONCLUSION

Hundreds of PGT-A studies are published worldwide, reporting on WCA, SA, WCM, and SM. Although these studies show agreement in some areas, such as WCA statistics and age dependence, they diverge on other issues, especially mosaicism and SA estimates. These disagreements are not trivial technical issues, and they impact two fundamental questions: whether and to what extent human embryos can undergo genetic self-correction and whether and under what circumstances mosaic or aneuploid embryos can be transferred. To address these issues effectively, one should critically analyze accumulated data to separate facts from artifacts and speculation. This critical review of published literature aims to outline evidence-based limits for genetic aberrations.

Current genetic aberration statistics are as follows: WCAs are predominantly maternal and strongly age-dependent. In oocytes, they affect roughly 20%–30% of eggs in women aged <35 years and increase by approximately 6% per year thereafter. Sperm WCAs are likely independent of age and other factors (except specific genetic cases) and affect 2%–3% of sperm cells. Embryonic WCAs are 80%–95% maternal, depending on age, with 50%–70% arising from oocyte meiosis I, 20%–30% from meiosis II, and 5%–15% from errors in the first mitosis. These manifest as aneuploid-aneuploid mosaics, which are often not distinguished from fully aneuploid embryos in clinical practice.

Segmental aneuploidies and mosaicism are largely independent of age and WCA. Segmental aneuploidy estimates vary widely (5%–30%) across studies due to artifacts from single cell analysis. Meiotic SAs are best characterized, showing a clear 2-phase developmental dynamic: 3%–10% in oocytes; 5%–20% in spermatozoa; 15%–25% in cleavage-stage embryos; and 7–15% in blastocysts. The increase from the oocyte to cleavage-stage is due to sperm SA, whereas the decrease from cleavage to blastocyst is likely linked to the reduced developmental potential of SA-affected embryos. Reported mosaicism rates also vary significantly (5%–100%), with most being artifacts. True biologic mosaicism likely affects 5%–15% of blastocysts; however, its dynamics in embryo development remain unclear due to poor-quality data from earlier stages and plenty of method-specific artifacts. The potential embryo self-correction fits within this uncertainty, although direct evidence is still emerging and requires further high-quality investigation.

Acknowledgments

The authors thank Alexei Slepzof and Sergey Popov for valuable discussions on molecular and bioinformatics issues.

CRediT Authorship Contribution Statement

Ilya Volodyaev: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Anna Ivanova:** Writing – review & editing, Writing – original draft, Visualization, Data curation. **Elena Korchivaia:** Writing – review & editing, Writing – original draft, Visualization, Data curation. **Alexey Surnov:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Ekaterina Pomerantseva:** Writing – review & editing. **Igor N. Lebedev:** Writing – review & editing. **Maria L. Semenova:** Writing – review & editing. **Ilya Mazunin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of Interests

I.V. is a board member of Interdisciplinary Clinical Association for Reproductive Medicine, a nonprofit company promoting knowledge exchange in the field of modern reproductive technologies and their introduction into clinical practice. A.I. has nothing to disclose. E.K. has nothing to disclose. A.S. has nothing to disclose. E.P. has nothing to disclose. I.N.L. has nothing to disclose. M.L.S. has nothing to disclose. I.M. is a board member of Interdisciplinary Clinical Association for Reproductive Medicine, a nonprofit company promoting knowledge exchange in the field of modern reproductive technologies and their introduction into clinical practice.

REFERENCES

1. Katagiri Y, Jwa SC, Kuwahara A, Iwasa T, Ono M, Kato K, et al. Assisted reproductive technology in Japan: a summary report for 2019 by the Ethics Committee of the Japan Society of Obstetrics and Gynecology. Reprod Med Biol 2022;21:e12434.
2. Liu K, Case A. Reproductive Endocrinology and Infertility Committee. Advanced reproductive age and fertility. J Obstet Gynaecol Can 2011;33:1165–75.
3. Practice Committee of the American Society for Reproductive Medicine and the Practice Committee of the Society for Reproductive Endocrinology

and Infertility. Optimizing natural fertility: a committee opinion. Fertil Steril 2022;117:53–63.
4. Viotti M. Preimplantation genetic testing for chromosomal abnormalities: aneuploidy, mosaicism, and structural rearrangements. Genes (Basel) 2020;11:602.
5. Waldvogel SM, Posey JE, Goodell MA. Human embryonic genetic mosaicism and its effects on development and disease. Nat Rev Genet 2024; 25:698–714.
6. Vollger MR, Dishuck PC, Harvey WT, DeWitt WS, Guitart X, Goldberg ME, et al. Increased mutation and gene conversion within human segmental duplications. Nature 2023;617:325–34.
7. Musson R, Gąsior Ł, Bisogno S, Ptak GE. DNA damage in preimplantation embryos and gametes: specification, clinical relevance and repair strategies. Hum Reprod Update 2022;28:376–99.
8. Xie P, Liu P, Zhang S, Cheng D, Chen D, Tan YQ, et al. Segmental aneuploidies with 1 Mb resolution in human preimplantation blastocysts. Genet Med 2022;24:2285–95.
9. Picchetta L, Ottolini CS, O'Neill HC, Capalbo A. Investigating the significance of segmental aneuploidy findings in preimplantation embryos. F S Sci 2023;4:17–26.
10. Marin D, Xu J, Treff NR. Preimplantation genetic testing for aneuploidy: a review of published blastocyst reanalysis concordance data. Prenat Diagn 2021;41:545–53.
11. Viotti M, Greco E, Grifo JA, Madjunkov M, Librach C, Cetinkaya M, et al. Chromosomal, gestational, and neonatal outcomes of embryos classified as a mosaic by preimplantation genetic testing for aneuploidy. Fertil Steril 2023;120:957–66.
12. Victor AR, Tyndall JC, Brake AJ, Lepkowsky LT, Murphy AE, Griffin DK, et al. One hundred mosaic embryos transferred prospectively in a single clinic: exploring when and why they result in healthy pregnancies. Fertil Steril 2019;111:280–93.
13. Barad DH, Albertini DF, Molinari E, Gleicher N. IVF outcomes of embryos with abnormal PGT-A biopsy previously refused transfer: a prospective cohort study. Hum Reprod 2022;37:1194–206.
14. Gleicher N, Patrizio P, Mochizuki L, Barad DH. Previously reported and here added cases demonstrate euploid pregnancies followed by PGT-A as "mosaic" as well as "aneuploid" designated embryos. Reprod Biol Endocrinol 2023;21:25.
15. Lin J, Vitek W, Scott EL. Order from chaos: a case report of a healthy live birth from a genetically "chaotic" embryo. F S Rep 2022;3:301–4.
16. Rodrigo L, Campus Galindo I, Polo Picasso A, Castello Salom D, Lopez de Carvajal LM, Simon Valles C. Embryo rebiopsy as a rescue tool in PGT-A cycles. 2021 ASEBIR Conference Toledo, Spain, 2021.
17. Campos G. PGT-A mosaicism based on NGS intermediate copy numbers: is it time to stop reporting them? J Assist Reprod Genet 2023;40:2925–32.
18. Capalbo A, Rienzi L. Mosaicism between trophectoderm and inner cell mass. Fertil Steril 2017;107:1098–106.
19. Capalbo A, Ubaldi FM, Rienzi L, Scott R, Treff N. Detecting mosaicism in trophectoderm biopsies: current challenges and future possibilities. Hum Reprod 2017;32:492–8.
20. Swain JE. Controversies in ART: can the IVF laboratory influence preimplantation embryo aneuploidy? Reprod Biomed Online 2019;39:599–607.
21. Gleicher N, Metzger J, Croft G, Kushnir VA, Albertini DF, Barad DH. A single trophectoderm biopsy at blastocyst stage is mathematically unable to determine embryo ploidy accurately enough for clinical use. Reprod Biol Endocrinol 2017;15:33.
22. Gleicher N, Patrizio P, Brivanlou A. Preimplantation genetic testing for aneuploidy-a castle built on sand. Trends Mol Med 2021;27:731–42.
23. Gleicher N, Albertini DF, Patrizio P, Orvieto R, Adashi EY. The uncertain science of preimplantation and prenatal genetic testing. Nat Med 2022;28:442–4.
24. Tiegs AW, Tao X, Zhan Y, Whitehead C, Kim J, Hanson B, et al. A multicenter, prospective, blinded, nonselection study evaluating the predictive value of an aneuploid diagnosis using a targeted next-generation sequencing-based preimplantation genetic testing for aneuploidy assay and impact of biopsy. Fertil Steril 2021;115:627–37.
25. Klimczak AM, Reig A, Neal SA, Seli E, Scott Jr RT. Interpretation of noninvasive prenatal testing results following in vitro fertilization and preimplantation genetic testing for aneuploidy. Am J Obstet Gynecol MFM 2020;2: 100232.
26. Franasiak JM, Forman EJ, Hong KH, Werner MD, Upham KM, Treff NR, et al. Aneuploidy across individual chromosomes at the embryonic level in trophectoderm biopsies: changes with patient age and chromosome structure. J Assist Reprod Genet 2014;31:1501–9.
27. Centers for Disease Control and Prevention. 2021 assisted reproductive technology (ART) fertility clinic and national summary report. Available at: https://www.cdc.gov/art/reports/2021/index.html. Accessed December 5, 2024.
28. Ishihara O, Jwa SC, Kuwahara A, Katagiri Y, Kuwabara Y, Hamatani T, et al. Assisted reproductive technology in Japan: a summary report for 2017 by the Ethics Committee of the Japan Society of Obstetrics and Gynecology. Reprod Med Biol 2020;19:3–12.
29. Practice Committees of the American Society for Reproductive Medicine and the Society for Assisted Reproductive Technology. The use of preimplantation genetic testing for aneuploidy (PGT-A): a committee opinion. Fertil Steril 2018;109:429–36.
30. Carrasquillo RJ, Kohn TP, Cinnioglu C, Rubio C, Simon C, Ramasamy R, et al. Advanced paternal age does not affect embryo aneuploidy following blastocyst biopsy in egg donor cycles. J Assist Reprod Genet 2019;36: 2039–45.
31. Fonseka KG, Griffin DK. Is there a paternal age effect for aneuploidy? Cytogenet Genome Res 2011;133:280–91.
32. Gao J, Wei N, Zhu X, Li R, Yan L, Qiao J. The correlation between morphological parameters and the incidence of de novo chromosomal abnormalities in 3238 biopsied blastocysts. J Assist Reprod Genet 2023;40:1089–98.
33. McCoy RC, Demko ZP, Ryan A, Banjevic M, Hill M, Sigurjonsson S, et al. Evidence of selection against complex mitotic-origin aneuploidy during preimplantation development. PLoS Genet 2015;11:e1005601.
34. Practice Committees of the American Society for Reproductive Medicine and the Society for Assisted Reproductive Technology. The use of preimplantation genetic testing for aneuploidy: a committee opinion. Fertil Steril 2024;122:421–34.
35. Kubicek D, Hornak M, Horak J, Navratil R, Tauwinklova G, Rubes J, et al. Incidence and origin of meiotic whole and segmental chromosomal aneuploidies detected by karyomapping. Reprod Biomed Online 2019;38:330–9.
36. Rabinowitz M, Ryan A, Gemelos G, Hill M, Baner J, Cinnioglu C, et al. Origins and rates of aneuploidy in human blastomeres. Fertil Steril 2012;97: 395–401.
37. Bell AD, Mello CJ, Nemesh J, Brumbaugh SA, Wysoker A, McCarroll SA. Insights into variation in meiosis from 31,228 human sperm genomes. Nature 2020;583:259–64.
38. Armstrong A, Kroener L, Miller J, Nguyen A, Kwan L, Quinn M. The nature of embryonic mosaicism across female age spectrum: an analysis of 21,345 preimplantation genetic testing for aneuploidy cycles. F S Rep 2023;4:256–61.
39. Ai X, Shi Y, Liu LW, Xu Y, Zhang H, Liu Y, et al. Risk factors related to chromosomal mosaicism in human blastocysts. Reprod Biomed Online 2022; 45:54–62.
40. Lebedev IN, Zhigalina DI. From contemplation to classification of chromosomal mosaicism in human preimplantation embryos. J Assist Reprod Genet 2021;38:2833–48.
41. Coticchio G, Barrie A, Lagalla C, Borini A, Fishel S, Griffin D, et al. Plasticity of the human preimplantation embryo: developmental dogmas, variations on themes and self-correction. Hum Reprod Update 2021;27:848–65.
42. Tšuiko O, Vanneste M, Melotte C, Ding J, Debrock S, Masset H, et al. Haplotyping-based preimplantation genetic testing reveals parent-of-origin specific mechanisms of aneuploidy formation. NPJ Genom Med 2021;6:81.
43. Wartosch L, Schindler K, Schuh M, Gruhn JR, Hoffmann ER, McCoy RC, et al. Origins and mechanisms leading to aneuploidy in human eggs. Prenat Diagn 2021;41:620–30.
44. McCarty KJ, Haywood ME, Lee R, Henry L, Arnold A, McReynolds S, et al. Segmental aneuploid hotspots identified across the genome concordant on reanalysis. Mol Hum Reprod 2022;29:gaac040.
45. Babariya D, Fragouli E, Alfarawati S, Spath K, Wells D. The incidence and origin of segmental aneuploidy in human oocytes and preimplantation embryos. Hum Reprod 2017;32:2549–60.

46. Navratil R, Horak J, Hornak M, Kubicek D, Balcova M, Tauwinklova G, et al. Concordance of various chromosomal errors among different parts of the embryo and the value of re-biopsy in embryos with segmental aneuploidies. Mol Hum Reprod 2020;26:269–76.
47. Victor AR, Griffin DK, Brake AJ, Tyndall JC, Murphy AE, Lepkowsky LT, et al. Assessment of aneuploidy concordance between clinical trophectoderm biopsy and blastocyst. Hum Reprod 2019;34:181–92.
48. Girardi L, Serdarogullari M, Patassini C, Poli M, Fabiani M, Caroselli S, et al. Incidence, origin, and predictive model for the detection and clinical management of segmental aneuploidies in human embryos. Am J Hum Genet 2020;106:525–34.
49. Walters-Sen L, Neitzel D, Bristow SL, Mitchell A, Alouf CA, Aradhya S, et al. Experience analysing over 190,000 embryo trophectoderm biopsies using a novel FAST-SeqS preimplantation genetic testing assay. Reprod Biomed Online 2022;44:228–38.
50. Fragouli E, Alfarawati S, Spath K, Jaroudi S, Sarasa J, Enciso M, et al. The origin and impact of embryonic aneuploidy. Hum Genet 2013;132:1001–13.
51. Magli MC, Albanese C, Crippa A, Terzuoli G, La Sala G, Tabanelli C, et al. Permanence of de novo segmental aneuploidy in sequential embryo biopsies. Hum Reprod 2020;35:759–69.
52. Coticchio G, Cimadomo D, Cermisoni GC, Rienzi L, Papaleo E, Ubaldi FM, et al. The first mitotic division: a perilous bridge connecting the zygote and the early embryo. Hum Reprod 2023;38:1019–27.
53. Popovic M, Dhaenens L, Boel A, Menten B, Heindryckx B. Chromosomal mosaicism in human blastocysts: the ultimate diagnostic dilemma. Hum Reprod Update 2020;26:313–34.
54. Popovic M, Borot L, Lorenzon AR, Lopes ALRC, Sakkas D, Lledó B, et al. Implicit bias in diagnosing mosaicism amongst preimplantation genetic testing providers: results from a multicenter study of 36 395 blastocysts. Hum Reprod 2024;39:258–74.
55. Martin RH. Meiotic errors in human oogenesis and spermatogenesis. Reprod Biomed Online 2008;16:523–31.
56. Vega M, Breborowicz A, Sauerbrun M, Stein D, Mcgovern P, Keltz M. Day 3 biopsy and blastulation rates. J Reprod Med 2016;61:336–40.
57. Zilberberg E, Casper R, Meriano J, Barzilay E, Aizer A, Kirshenbaum M, et al. Cleavage vs blastocyst stage embryos: how are they interrelating? Arch Gynecol Obstet 2021;304:1083–8.
58. Escribà MJ, Vendrell X, Peinado V. Segmental aneuploidy in human blastocysts: a qualitative and quantitative overview. Reprod Biol Endocrinol 2019;17:76.
59. Kim J, Tao X, Cheng M, Steward A, Guo V, Zhan Y, et al. The concordance rates of an initial trophectoderm biopsy with the rest of the embryo using PGTseq, a targeted next-generation sequencing platform for preimplantation genetic testing-aneuploidy. Fertil Steril 2022;117:315–23.
60. Lebedev IN, Shilova NV, Yurov IY, Malysheva OV, Tveleneva AA, Minzhenkova ME, et al. Guidelines of the Russian Society of Medical Geneticists for Chromosomal Microarray Analysis. Med Genet 2023;22:3–47.
61. Katz-Jaffe M, McReynolds S, Klerk K, Henry LN, Schweitz M, Swain J, et al. Extremely low incidence of mosaicism in human blastocysts mimics occurrence in natural and IVF clinical pregnancies. Fertil Steril 2017;108:e87–8.
62. Colorado Center for Reproductive Medicine. CCRM reports the lowest chromosomal mosaicism rates in the U.S. at less than three percent across the CCRM Network. Available at: https://www.ccrmivf.com/news-events/ccrm-lowest-chromosomal-mosaicism-rates-usa/. Accessed December 5, 2024.
63. Escudero T, Ribustello L, Sumarroca M, Munne S. Next generation sequencing (NGS) and the rate of partial aneuploidy in preimplantation genetic screening (PGS). Fertil Steril 2016;105:e20–1.
64. Munné S, Grifo J, Cohen J, Weier HU. Chromosome abnormalities in human arrested preimplantation embryos: a multiple-probe FISH study. Am J Hum Genet 1994;55:150–9.
65. Mccoy RC, Summers MC, Mccollin A, Ottolini CS, Ahuja K, Handyside AH. Meiotic and mitotic aneuploidies drive arrest of in vitro fertilized human preimplantation embryos. Genome Med 2023;15:77.
66. Daphnis DD, Fragouli E, Economou K, Jerkovic S, Craft IL, Delhanty JD, et al. Analysis of the evolution of chromosome abnormalities in human embryos from Day 3 to 5 using CGH and FISH. Mol Hum Reprod 2007;14:117–25.
67. Gonzalez-Merino E, Emiliani S, Vassart G, Van den Bergh M, Vannin AS, Abramowicz M, et al. Incidence of chromosomal mosaicism in human embryos at different developmental stages analyzed by fluorescence in situ hybridization. Genet Test 2003;7:85–95.
68. Belyaev A, Tofilo M, Popov S, Mazunin I, Fomin D. The assessment of methods for preimplantation genetic testing for aneuploidies using a universal parameter: implications for costs and mosaicism detection. J Biotechnol Biomed 2024;7:47–59.
69. Munné S, Grifo J, Wells D. Mosaicism: "survival of the fittest" versus "no embryo left behind". Fertil Steril 2016;105:1146–9.
70. Chan CW, Lee CSS, Yap WY, Lim YX. Impact of maternal age on mosaicism rate in preimplantation embryos. A retrospective study. Reprod Biomed Online 2019;38:e39–40.
71. Steuerwald N, Cohen J, Herrera RJ, Sandalinas M, Brenner CA. Association between spindle assembly checkpoint expression and maternal age in human oocytes. Mol Hum Reprod 2001;7:49–55.
72. Brooks KE, Daughtry BL, Davis B, Yan MY, Fei SS, Shepherd S, et al. Molecular contribution to embryonic aneuploidy and karyotypic complexity in initial cleavage divisions of mammalian development. Development 2022;149:dev198341.
73. Pavin N, Tolić IM. Mechanobiology of the mitotic spindle. Dev Cell 2021;56:192–201.
74. Ong JY, Bradley MC, Torres JZ. Phospho-regulation of mitotic spindle assembly. Cytoskeleton (Hoboken) 2020;77:558–78.
75. Ford E, Currie CE, Taylor DM, Erent M, Marston AL, Hartshorne GM, et al. The first mitotic division of the human embryo is highly error-prone. Preprint. Posted online July 17, 2020. bioRxiv 2020:2020.07.17.208744. https://doi.org/10.1101/2020.07.17.208744
76. Kai Y, Kawano H, Yamashita N. First mitotic spindle formation is led by sperm centrosome-dependent MTOCs in humans. Reproduction 2021;161:V19–22.
77. Scott RT Jr, Upham KM, Forman EJ, Zhao T, Treff NR. Cleavage-stage biopsy significantly impairs human embryonic implantation potential while blastocyst biopsy does not: a randomized and paired clinical trial. Fertil Steril 2013;100:624–30.
78. Treff NR, Levy B, Su J, Northrop LE, Tao X, Scott Jr RT. SNP microarray-based 24 chromosome aneuploidy screening is significantly more consistent than FISH. Mol Hum Reprod 2010;16:583–9.
79. Velilla E, Escudero T, Munné S. Blastomere fixation techniques and risk of misdiagnosis for preimplantation genetic diagnosis of aneuploidy. Reprod Biomed Online 2002;4:210–7.
80. Chuang TH, Chang YP, Lee MJ, Wang HL, Lai HH, Chen SU. The incidence of mosaicism for individual chromosome in human blastocysts is correlated with chromosome length. Front Genet 2021;11:565348.
81. Bielanska M, Tan SL, Ao A. Chromosomal mosaicism throughout human preimplantation development in vitro: incidence, type, and relevance to embryo outcome. Hum Reprod 2002;17:413–9.
82. Baart EB, Martini E, van den Berg I, Macklon NS, Galjaard RJ, Fauser BC, et al. Preimplantation genetic screening reveals a high incidence of aneuploidy and mosaicism in embryos from young women undergoing IVF. Hum Reprod 2006;21:223–33.
83. Viotti M, Victor AR, Barnes FL, Zouves CG, Besser AG, Grifo JA, et al. Using outcome data from one thousand mosaic embryo transfers to formulate an embryo ranking system for clinical use. Fertil Steril 2021;115:1212–24.
84. Zhang YX, Chen JJ, Nabu S, Yeung QSY, Li Y, Tan JH, et al. The pregnancy outcome of mosaic embryo transfer: a prospective multicenter study and meta-analysis. Genes (Basel) 2020;11:973.
85. Yakovlev P, Vyatkina S, Polyakov A, Pavlova M, Volkomorov V, Yakovlev M, et al. Neonatal and clinical outcomes after transfer of a mosaic embryo identified by preimplantation genetic testing for aneuploidies. Reprod Biomed Online 2022;45:88–100.
86. Greco E, Minasi MG, Fiorentino F. Healthy babies after intrauterine transfer of mosaic aneuploid blastocysts. N Engl J Med 2015;373:2089–90.
87. Capalbo A, Poli M, Rienzi L, Girardi L, Patassini C, Fabiani M, et al. Mosaic human preimplantation embryos and their developmental potential in a prospective, non-selection clinical trial. Am J Hum Genet 2021;108:2238–47.

88. Spinella F, Fiorentino F, Biricik A, Bono S, Ruberti A, Cotroneo E, et al. Extent of chromosomal mosaicism influences the clinical outcome of in vitro fertilization treatments. Fertil Steril 2018;109:77–83.
89. Munné S, Blazek J, Large M, Martinez-Ortiz PA, Nisson H, Liu E, et al. Detailed investigation into the cytogenetic constitution and pregnancy outcome of replacing mosaic blastocysts detected with the use of high-resolution next-generation sequencing. Fertil Steril 2017;108:62–71.e8.
90. Mourad A, Antaki R, Bissonnette F, Al Baini O, Saadeh B, Jamal W. Evidence-based clinical prioritization of embryos with mosaic results: a systematic review and meta-analysis. J Assist Reprod Genet 2021;38:2849–60.
91. Munné S, Spinella F, Grifo J, Zhang J, Beltran MP, Fragouli E, et al. Clinical outcomes after the transfer of blastocysts characterized as mosaic by high resolution next generation sequencing- further insights. Eur J Med Genet 2020;63:103741.
92. Campbell A, Fishel S, Bowman N, Duffy S, Sedler M, Hickman CF. Modelling a risk classification of aneuploidy in human embryos using non-invasive morphokinetics. Reprod Biomed Online 2013;26:477–85.
93. Qi ST, Liang LF, Xian YX, Liu JQ, Wang W. Arrested human embryos are more likely to have abnormal chromosomes than developing embryos from women of advanced maternal age. J Ovarian Res 2014;7:65.
94. Vega M, Breborowicz A, Moshier EL, McGovern PG, Keltz MD. Blastulation rates decline in a linear fashion from euploid to aneuploid embryos with single versus multiple chromosomal errors. Fertil Steril 2014;102:394–8.
95. Bolton H, Graham SJL, Van der Aa N, Kumar P, Theunis K, Fernandez Gallardo E, et al. Mouse model of chromosome mosaicism reveals lineage-specific depletion of aneuploid cells and normal developmental potential. Nat Commun 2016;7:11165.
96. Yang M, Rito T, Metzger J, Naftaly J, Soman R, Hu J, et al. Depletion of aneuploid cells in human embryos and gastruloids. Nat Cell Biol 2021;23:314–21.
97. Griffin DK, Brezina PR, Tobler K, Zhao Y, Silvestri G, Mccoy RC, et al. The human embryonic genome is karyotypically complex, with chromosomally abnormal cells preferentially located away from the developing fetus. Hum Reprod 2023;38:180–8.
98. Coticchio G, Lagalla C, Sturmey R, Pennetta F, Borini A. The enigmatic morula: mechanisms of development, cell fate determination, self-correction and implications for ART. Hum Reprod Update 2019;25:422–38.
99. Sermon K, Capalbo A, Cohen J, Coonen E, De Rycke M, De Vos A, et al. The why, the how and the when of PGS 2.0: current practices and expert opinions of fertility specialists, molecular biologists, and embryologists. Mol Hum Reprod 2016;22:845–57.
100. Leigh D, Cram DS, Rechitsky S, Handyside A, Wells D, Munne S, et al. PGDIS position statement on the transfer of mosaic embryos 2021. Reprod Biomed Online 2022;45:19–25.
101. Preimplantation Genetic Diagnosis International Society. PGDIS position statement on chromosome mosaicism and preimplantation aneuploidy testing at the blastocyst stage. PGDIS Newsletter; 2016.
102. Practice Committee and Genetic Counseling Professional Group (GCPG) of the American Society for Reproductive Medicine. Clinical management of mosaic results from preimplantation genetic testing for aneuploidy (PGT-A) of blastocysts: a committee opinion. Fertil Steril 2020;114:246–54.
103. Illumina. Single cell washing and handling. Available at: https://www.mountsinaiservices.com/wp-content/uploads/2017/12/Single-Cell-Washing-and-Handling-15056987-A.pdf. Accessed December 5, 2024.
104. Illumina. A technical guide to aneuploidy calling with VeriSeq PGS. Available at: https://support.illumina.com/content/dam/illumina-support/documents/documentation/chemistry_documentation/veriseq-pgs/veriseq-pgs-technical-guide-to-aneuploidy-calling-15059470-a.pdf. Accessed December 5, 2024.
105. Sims D, Sudbery I, Ilott NE, Heger A, Ponting CP. Sequencing depth and coverage: key considerations in genomic analyses. Nat Rev Genet 2014;15:121–32.
106. Ghevaria H, SenGupta S, Naja R, Odia R, Exeter H, Serhal P, et al. Next generation sequencing detects premeiotic errors in human oocytes. Int J Mol Sci 2022;23:665.
107. Sabater MR, Lledo B, Ortiz JA, Cascales A, Ten J, Bernabeu A, et al. P-519 Relationship between the quality control parameters in PGT-A results by NGS with the chromosomal mosaicism diagnosis in trophoectoderm biopsies. Hum Reprod 2022;37:deac107.481.
108. Illumina. VeriSeq PGS guidance note. Available at: https://support.illumina.com/content/dam/illumina-support/documents/documentation/chemistry_documentation/veriseq-pgs/veriseq-pgs-guidance-note-1000000138837-00.pdf. Accessed December 5, 2024.
109. Bouwman AC, Hulsegge I, Hawken RJ, Henshall JM, Veerkamp RF, Schokker D, et al. Classifying aneuploidy in genotype intensity data using deep learning. J Anim Breed Genet 2023;140:304–15.
110. Bakker B, Taudt A, Belderbos ME, Porubsky D, Spierings DC, de Jong TV, et al. Single-cell sequencing reveals karyotype heterogeneity in murine and human malignancies. Genome Biol 2016;17:115.